

A 4B model watches the video and raises the alert itself.

Real-time video anomaly detection · 11 classes
Qwen3-VL-4B-Instruct · one A100-40GB
34 public test videos · 56 minutes of footage

Every 10 seconds a new 20-second window closes. 16 frames go to the model; it answers with a class, a confidence and a sentence of reasoning.

INCOMING VIDEO



TIME BUDGET FOR ONE WINDOW



75 / 100

marks had no training data

Train ships 5–30 s clips with one event. D2 and D3 are 4–10 minute videos with several. We synthesised 235, and deleted 125 more whose timestamps we could not verify.

5.5×

faster than real time

1.83 s to score a window that arrives every 10 s. Still 2.8× at the 95th percentile, so it holds on the tail.

4B

or 8B — both self-hosted

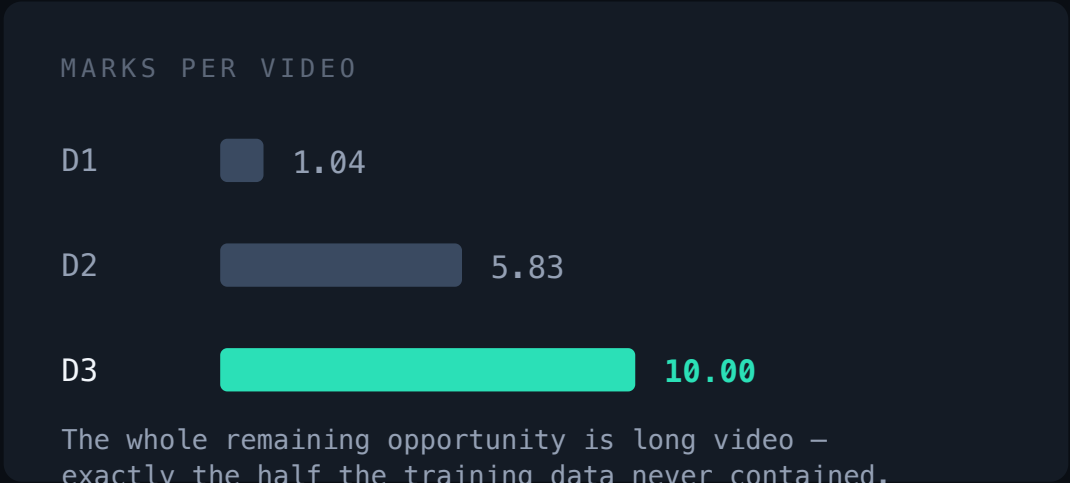
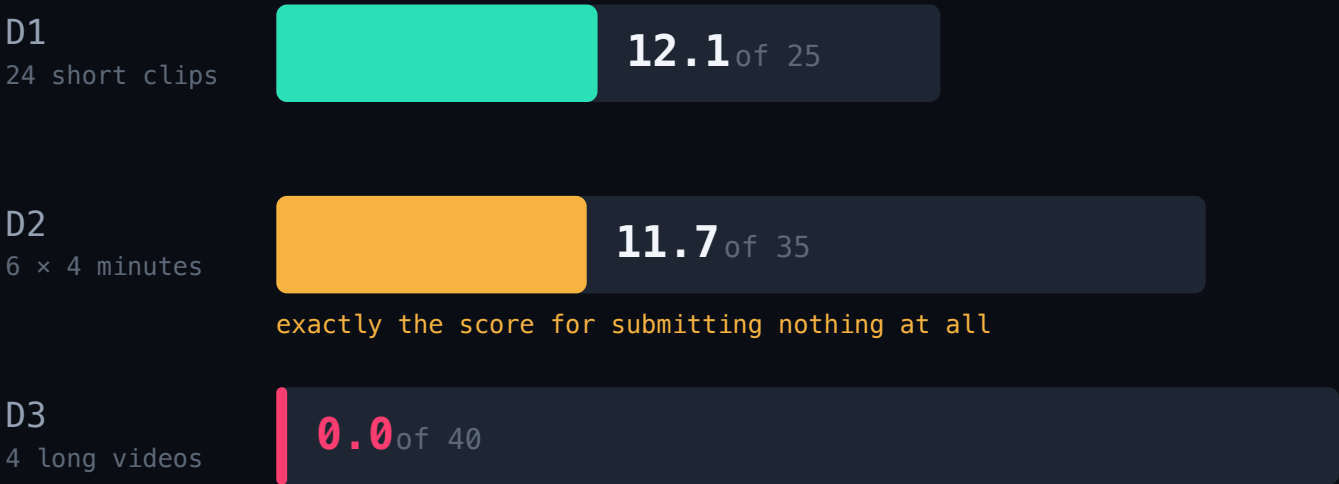
Indistinguishable on the practice set (3 videos of 24, $p = 0.25$). On the private eval pack the 8B produced our best submission. Nothing hosted is in the runtime path.

Demo: a dashboard runs this live — press a button, one window goes to the A100, the alert comes back in about nine seconds.

Short clips work. Long video does not — and that is where the marks are.

First scored upload · zero-shot, no fine-tuning yet
23.8 / 100
24 videos, 20 positives → ±1 detection ≈ 5 pp of recall

Marks earned against marks available. One D3 video is worth ten D1 videos.



12 → 3

false alarms, for free

Keeping only the top answer per clip cut them 4x with no loss of recall, on CPU. The model's own confidence turned out to be useless as a filter.

7

silent failures, 0 crashes

Every one would have read as “the model isn't very good” — and the fix for the fifth caused the seventh. So we validate intermediate artifacts, not outputs.

0.556

AUC where chance is 0.500

The ego-motion gate every fixed-camera paper implies. We tested it against a permutation null: it does not work. A day of GPU saved.